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# Mathematical modeling of Chinese medicine by complex-valued five-agent network

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**Abstract**
Traditional Chinese medicine (TCM) is a holistic medicine that deals with the whole person, not just the disease or injury. Based on the Yin-Yang Philosophy and the Wuxing (Five Elements) network, TCM advocates that the human body has the self-regulation ability, which can make the internal organs of the human body operate in a coordinated manner, adapt to changes in the external environment, and achieve harmony between the human body and nature. However, for more than two thousand years, this argument about the human body's self-regulation and adaptation abilities has remained at the level of natural philosophy, lacking direct scientific evidence. This article applies the complex-valued multi-agent network theory to build a general mathematical model for the Yin-Yang Wuxing network based on which, we can quantitatively describe and compute the operation of TCM by linear algebra. The eigenstructure analysis of the established mathematical model clearly reveals the roles of the Yin-Yang philosophy and the Wuxing network in TCM and confirms their inherent feedback mechanism to ensure the balanced operation of organs in the human body. This paper opens the way for quantitative research on TCM, which can be used to identify unbalanced organs and predict the impact of environmental changes and medical treatments on the human body.

**Highlights**

- Establish a mathematical platform based on complex-valued five-agent network for studying TCM in a quantitative and computable way.
- Analyze the cooperative and competitive interactions in TCM to reveal the human body's self-regulation and adaptation abilities
- Predict the impact of environmental input and medical treatment on the human body through the simulation of the proposed model of TCM.

**Keywords** Traditional Chinese medicine · Mathematical modeling · Complex-valued five-agent network · Yin-Yang philosophy

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## 1 Introduction

The Yin-Yang philosophy and the Wuxing (Five Elements in Chinese) network have been the two major philosophical foundations of traditional Chinese medicine (TCM) for more than two thousand years [1, 2]. However, because these two philosophical cornerstones cannot be quantitatively described in mathematical language, TCM still lacks connection with modern science. Based on the advancements in modeling technology [3–5] and the recent breakthroughs in the scientific interpretation of the Yin-Yang philosophy and the Wuxing network, it has become feasible to establish a mathematical platform for analyzing and computing TCM. It was pointed out that the mathematical correspondence of the Yin-Yang complementarity is the real-imaginary complementarity [6], and the scientific counterpart of the Wuxing network (see Fig. 1a) is the five-agent network system [7]. Based on current findings, we can say that TCM constructed based on the Yin-Yang philosophy and the Wuxing network is mathematically equivalent to a five-agent network operating on the complex domain. This article aims to establish such a complex-valued five-agent network and analyze its eigenstructure to understand how organs in the human body achieve internal balance and how they adapt to environmental changes.

The origins and development of the Yin-Yang and Wuxing philosophical thoughts were originally independent. Their systematic integration first appeared in the Huangdi Neijing [8] (the *Inner Canon of the Yellow Emperor*), the ancestor of TCM. The Wuxing network, which matured in the second century BC to unify the laws of nature and human body, is the world's oldest five-element network and is represented pictographically by five natural substances: wood, fire, earth, metal, and water, arranged at the five vertices of a regular pentagon, as shown in Fig. 1a. After a lengthy period of evolution, the five elements in the Wuxing network no longer refer to five specific substances, but are abstracted into five properties, five steps, or five trends of the system [9]. From the classification of modern science, the ancient Wuxing

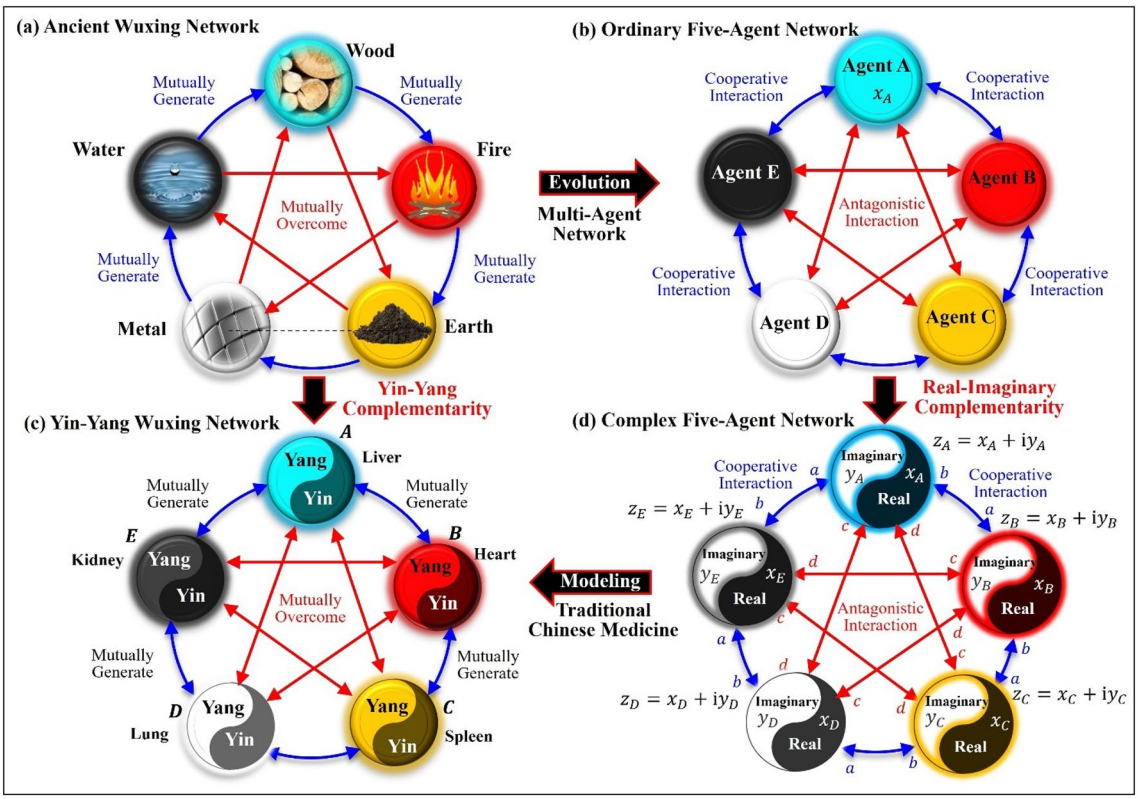



**Fig. 1** The evolution from the ancient Wuxing network to the complex five-agent network. **a** The ancient Wuxing network demonstrates the generating and overcoming interactions between the five natural substances. **b** The Wuxing network evolves into the ordinary five-agent network, whose operation is independent of the physical characteristics of the agents and is only related to their relative positions such that adjacent agents generate (cooperate with) each other and spaced-apart agents overcome (compete with) each other. **c** Adding the Yin-Yang philosophy to the Wuxing network forms the Yin-Yang Wuxing network, the main structure of TCM. **d** The ordinary five-agent network is complexified to yield the complex-valued five-agent network, which provides a mathematical model for TCM via the equivalence between the Yin-Yang complementarity and the real-imaginary complementarity



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network represents a multi-agent network [10, 11], whose number of agents is five, as shown in Fig. 1b. In the long-term historical evolution, the Wuxing network has evolved into diverse agents. For example, the agents of the Wuxing network in the human body are the five organs of liver, heart, spleen, lung, and kidney; the agents in the sky are the five planets of Jupiter, Mars, Saturn, Venus, and Mercury; the agents on the ground are the five seasons of spring, summer, long summer, autumn, and winter. It is based on the interactions among these three categories of agents that TCM explains the causes of human diseases and proposes the methods of medical treatment.

The operation of the ancient Wuxing network (see Fig. 1a) obeys the protocol that the adjacent substances generate (cooperate with) each other while the spaced-apart substances overcome (compete with) each other [12]. The ordinary five-agent network (see Fig. 1b) evolved from the Wuxing network is irrelevant to the physical entities represented by the agents as long as they obey the prescribed protocol. Under the simultaneous operation of cooperative and antagonistic interactions, the main issue of the ordinary five-agent network is how to maintain the stability of the network. It is interesting to find that the stability of the ordinary five-agent network is determined by the ratio of the cooperative strength to the antagonistic strength, and this ratio must be greater than the golden ratio [7], the world's most astonishing number originating from ancient Greece.

Another cornerstone of TCM is the thought of Yin and Yang [13, 14], which originated more than 3,000 years ago, even earlier than the Wuxing network. The Yin-Yang thought asserts that everything in the universe comprises Yin and Yang elements, called the complementary pairs, such as dark and bright, female and male, the moon and the sun, cold and hot, passive and active, and so on. Complementary pairs are opposite yet rely on each other to form a whole. Niels Bohr was the first person to introduce the principle of complementarity [15] into the world of physics and regarded it as a universal principle. Bohr designed his coat of arms [16] that featured a Yin-Yang pattern (Tai Chi symbol) to symbolize the wave-particle complementarity when he was made a knight of the Order of the Elephant by the King of Denmark in 1947.

The counterpart of Yin-Yang complementarity in physics is the wave-particle complementarity [17], while its counterpart in mathematics is the real-imaginary complementarity [6]. It was pointed out that the wave-particle complementarity directly results from quantum motion occurring in complex space [18, 19]. The validity of the complex-space interpretation of wave-particle complementarity is further supported by the recent findings of weak measurements, which reveal that every physical observable has a real part as well as an imaginary part [20, 21]. Like entanglement, imaginarity has been recognized as an indispensable resource responsible for quantum advantages [22–24], which ruled out the real-valued standard formalism of quantum theory [25].

As Niels Bohr said, the principle of complementarity is universal, and it is also true in TCM, where the Yin-Yang complementarity plays the same role as the wave-particle complementarity in physics. Just like modern physics interprets wave-particle complementarity in complex space, TCM uses a similar real-imaginary perspective to interpret Yin-Yang complementarity. The Huangdi Neijing, the oldest classic of TCM, mentions: "Yang transforms Qi, and Yin takes shape." In other words, Yang represents the vaporization of energy, whose attribute is invisible and imaginary, while Yin represents the condensation of matter, which is tangible and real. TCM explains the mutual transformation between matter and energy through the changes of Yin and Yang (see Fig. 1c), which can be expressed mathematically by real and imaginary numbers, respectively.

Because the operation of TCM is based on the Wuxing network and the Yin-Yang philosophy, its mathematical modeling consists of two parts: the modeling of the Wuxing network by the five-agent network and the modeling of the Yin-Yang philosophy by the real-imaginary complementarity. However, mathematical modeling of the Wuxing network and Yin-Yang philosophy has so far been conducted independently, lacking integration with each other. This paper combines the two independent models to yield the complex-valued five-agent network (see Fig. 1d), which can be used as a mathematical platform to quantitatively describe the operation of TCM and to explain how TCM maintains the balance of the human body and adapts to changes in the external environment.

## 2 Modeling TCM by complex-valued five-agent networks

According to TCM, the five agents in the Wuxing network correspond to the five organs in the human body: liver, heart, spleen, lungs, and kidney, as shown in Fig. 1c. Each organ has Yin component $x$, which represents the amount of tangible nutrients needed to maintain the life activities in the organ, such as carbohydrates, lipids, proteins, etc., as well as Yang



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component $y$, which represents the amount of invisible energy released by the oxidation of nutrients. Combining the Yin and Yang components, the five agents can be described quantitatively by five complex-valued indices as

$$\begin{bmatrix} z_A \\ z_B \\ z_C \\ z_D \\ z_E \end{bmatrix} = \begin{bmatrix} x_A \\ x_B \\ x_C \\ x_D \\ x_E \end{bmatrix} + i \begin{bmatrix} y_A \\ y_B \\ y_C \\ y_D \\ y_E \end{bmatrix}, \tag{1}$$

where the real part $x_i$ and the imaginary part $y_i$ denote the measures of the Yin component and the Yang component of the agent, respectively. The complexification process in Eq. (1) demonstrates the mathematical implication of the Yin-Yang complementarity that the real (Yin) and imaginary (Yang) components are opposed to each other but depend on each other to form a whole.

We note that there is no Yin-Yang distinction in the ancient Wuxing network (see Fig. 1a) because it represents a network of Yin-Yang balance so that there is no need to distinguish between Yin and Yang. When the Wuxing network is subject to external disturbances, the agent's Yin and Yang components lose balance and separate from each other, resulting in the emergence of the Yin-Yang Wuxing network, as shown in Fig. 1c. The Yin-Yang duality of the Wuxing network brings historical debate regarding the interaction between agents. Taking agent A as an example, the agent adjacent to A is B, so there should be a cooperative relationship between A and B. However, since both A and B have Yin and Yang components, the cooperative relationship between them may be that the Yin component $x_A$ cooperates with the Yin component $x_B$, or it may be that the Yin component $x_A$ cooperates with the Yang component $y_B$.

An analogous situation occurs in the mutual competition between agents A and C. It may be that the Yin component $x_A$ competes with the Yin component $x_C$, or it may be that the Yin component $x_A$ competes with the Yang component $y_C$. We can summarize the above different combinations into the following two rules of interaction:

- Type-I interaction: Those with the same Yin-Yang attribute cooperate, while those with opposite attributes compete, such as Yin-Yin (Yang-Yang) cooperation and Yin-Yang competition, as shown in Fig. 2a and b.
- Type-II interaction: Those with opposite Yin-Yang attributes cooperate, while those with the same attribute compete, such as Yin-Yang cooperation and Yin-Yin (Yang-Yang) competition, as shown in Fig. 2c and d.

There is no unified view on the correctness of the above two types of interactions in TCM. Due to the lack of mathematical tools, TCM scholars for a long time have only analyzed the above two interactions based on their philosophical viewpoints. Here, we employ the correspondence between the Yin-Yang Wuxing network and the complex-valued five-agent network to quantitatively analyze the impact of the two types of Yin-Yang interactions on the balance and internal stability in TCM.

## 3 Type-I interaction: Yin-Yin cooperation and Yin-Yang competition

Figure 2 takes agent A as the central agent to illustrate its cooperative relationship with adjacent agents B and E and its antagonistic relationship with the spaced-apart agents C and D. Figure 2a shows the four effects on $x_A$ according to the type-I interaction:

- Generating action (GA) from agent E:

According to the rule of the type-I interaction, cooperative interaction occurs between agents with the same attribute, so the cooperative interaction between A and E is determined either by $x_A$ and $x_E$ or by $y_A$ and $y_E$. The change rate of $x_A$ caused by GA from agent E is given by

$$\dot{x}_A = a\left(x_E - x_A\right), \tag{2}$$

where $a > 0$ is the constant rate of GA.



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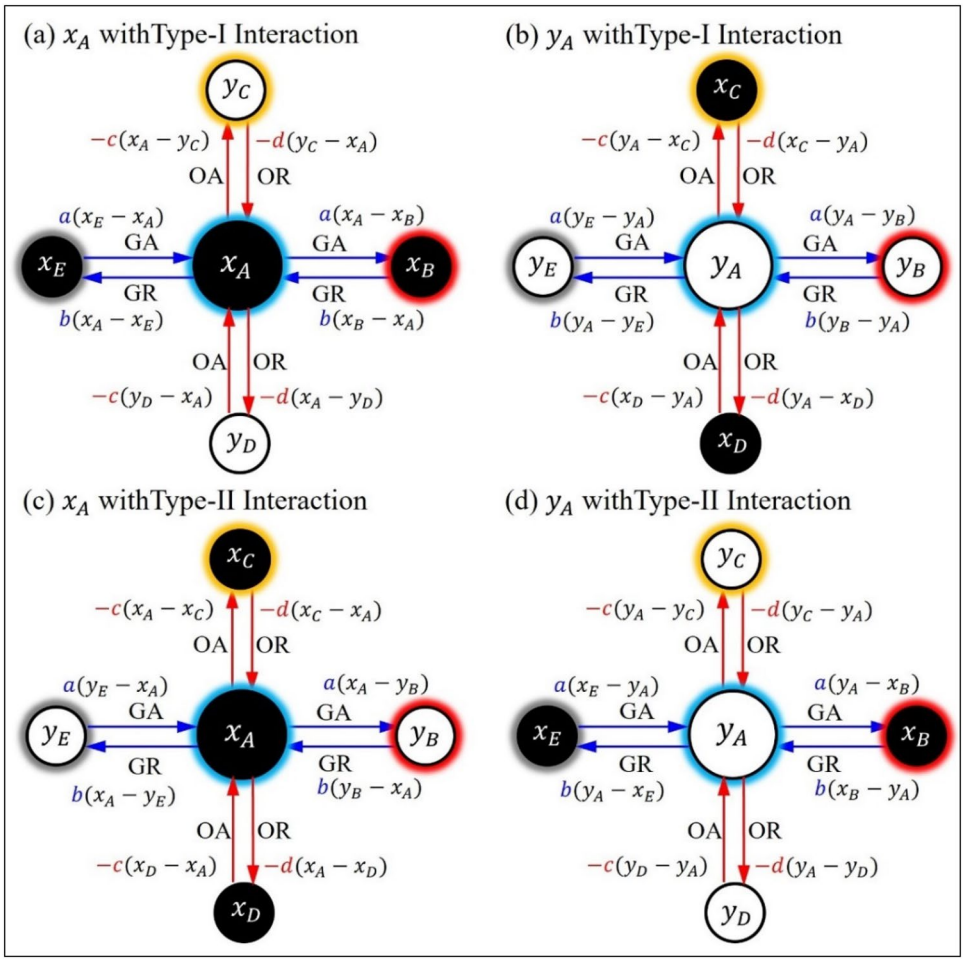



**Fig. 2** Two types of interaction between agents with Yin-Yang duality. Parts **a** and **b** illustrate type-I interaction that adjacent agents with the same attribute generate each other to produce generating action (GA) and generating reaction (GR), while spaced-apart agents with different attributes overcome each other to produce overcoming action (OA) and overcoming reaction (OR). Parts **c** and **d** illustrate the type-II interaction in which adjacent agents with different attributes generate each other, while spaced-apart agents with the same attribute overcome each other

- Generating reaction (GR) from agent B:

When agent A generates agent B, agent A also receives the reaction from agent B, and the resulting change rate of $x_A$ caused by GR is expressed by

$$\dot{x}_A = b(x_B - x_A), \tag{3}$$

where $b > 0$ is the constant rate of GR.

- Overcoming action (OA) from agent D:

According to the rule of the type-I interaction, overcoming interaction occurs between agents with different attributes, so the overcoming interaction between A and E is determined either by $x_A$ and $y_E$ or by $y_A$ and $x_E$. The change rate of $x_A$ caused by OA from agent D is then given by

$$\dot{x}_A = -c(y_D - x_A), \tag{4}$$

where $c > 0$ is the constant rate of OA.

- Overcoming reaction (OR) from agent C:

When agent A overcomes agent C, agent A also receives the reaction from agent C, and the resulting change rate of $x_A$ caused by OR is expressed by



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$$\dot{x}_A = -d(y_C - x_A), \tag{5}$$

where $d > 0$ is the constant rate of OR.

Adding the above four equations, we obtain the change rate of $x_A$ caused by the sum of the four effects:

$$\dot{x}_A = a(x_E - x_A) + b(x_B - x_A) - c(y_D - x_A) - d(y_C - x_A). \tag{6}$$

As to the Yang component of A, denoted by $y_A$, Fig. 2b shows the effects of the other four agents on $y_A$, where we can see that $y_E$ and $y_B$ have a generating interaction with $y_A$, while $x_D$ and $x_C$ have a overcoming interaction with $y_A$. Combining the two generating effects and the two overcoming effects on $y_A$ yields the total change rate of $y_A$ as

$$\dot{y}_A = a(y_E - y_A) + b(y_B - y_A) - c(x_D - y_A) - d(x_C - y_A). \tag{7}$$

Multiplying Eq. (7) by the imaginary number i and adding the resultant to Eq. (6), together with the complex notation $z_A = x_A + \mathrm{i}y_A$, we obtain the complex differential equation for $z_A$ as

$$\dot{z}_A = a(z_E - z_A) + b(z_B - z_A) - c(\mathrm{i}\bar{z}_D - z_A) - d(\mathrm{i}\bar{z}_C - z_A). \tag{8}$$

In the same way, we can derive the complex equations for the other four agents as follows:

$$\dot{z}_B = a(z_A - z_B) + b(z_C - z_B) - c(i\bar{z}_E - z_B) - d(\mathrm{i}\bar{z}_D - z_B), \tag{9}$$

$$\dot{z}_C = a(z_B - z_C) + b(z_D - z_C) - c(\mathrm{i}\bar{z}_A - z_C) - d(\mathrm{i}\bar{z}_E - z_C), \tag{10}$$

$$\dot{z}_D = a(z_C - z_D) + b(z_E - z_D) - c(\mathrm{i}\bar{z}_B - z_D) - d(\mathrm{i}\bar{z}_A - z_D), \tag{11}$$

$$\dot{z}_E = a(z_D - z_E) + b(z_A - z_E) - c(i\bar{z}_C - z_E) - d(i\bar{z}_B - z_E), \tag{12}$$

where the terms multiplied by the coefficients $a$ and $b$ are related to the generating interaction, while the terms multiplied by the coefficients $c$ and $d$ are related to the overcoming interaction. In addition, we note that the conjugate complex number $\bar{z}$ only appears in overcoming interactions. By separating the real and imaginary parts of the above five complex differential equations, we obtain the following matrix equation:

$$\begin{bmatrix} \dot{x}_A \\ \dot{y}_A \\ \dot{x}_B \\ \dot{y}_B \\ \dot{x}_C \\ \dot{y}_C \\ \dot{x}_D \\ \dot{y}_D \\ \dot{x}_E \\ \dot{y}_E \end{bmatrix} = \begin{bmatrix} \delta & 0 & b & 0 & 0 & -d & 0 & -c & a & 0 \\ 0 & \delta & 0 & b & -d & 0 & -c & 0 & 0 & a \\ a & 0 & \delta & 0 & b & 0 & 0 & -d & 0 & -c \\ 0 & a & 0 & \delta & 0 & b & -d & 0 & -c & 0 \\ 0 & -c & a & 0 & \delta & 0 & b & 0 & 0 & -d \\ -c & 0 & 0 & a & 0 & \delta & 0 & b & -d & 0 \\ 0 & -d & 0 & -c & a & 0 & \delta & 0 & b & 0 \\ -d & 0 & -c & 0 & 0 & a & 0 & \delta & 0 & b \\ b & 0 & 0 & -d & 0 & -c & a & 0 & \delta & 0 \\ 0 & b & -d & 0 & -c & 0 & 0 & a & 0 & \delta \end{bmatrix} \begin{bmatrix} x_A \\ y_A \\ x_B \\ y_B \\ x_C \\ y_C \\ x_D \\ y_D \\ x_E \\ y_E \end{bmatrix} \rightarrow \dot{\mathbf{X}}' = \mathbf{A}'\mathbf{X}', \tag{13}$$

where $\mathbf{X}' = \begin{bmatrix} x_A & y_A & x_B & y_B & x_C & y_C & x_D & y_D & x_E & y_E \end{bmatrix}^T$ is the state vector and the parameter $\delta$ is defined as $\delta = -(a + b - c - d)$. System matrix $\mathbf{A}'$ is a $10 \times 10$ square matrix, whose eigenvalues determines the stability of the complex five-agent network with the type-I interaction, and whose eigenvectors determine the grouping pattern of the Yin and Yang components. Except for $\lambda'_1 = 0$, the largest eigenvalue of $\mathbf{A}'$ is $\lambda'_{10} = 2(c + d) = 2c_+ > 0$, and the other eight eigenvalues are composed of four pairs of conjugate complex numbers, as listed in the second column of Table 1.



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**Table 1** Eigenvalues of four types of five-agent networks. The parameters are defined as $\alpha = (5-\sqrt{5})/2$, $\beta = (5+\sqrt{5})/2$, $a_{\pm} = a \pm b$, and $c_{\pm} = c \pm d$

| Ordinary five-agent network A | Complex five-agent network A′ with type-I interaction | Complex five-agent network A″ with type-II interaction | Complex five-agent network A‴ with self-generation |
|---|---|---|---|
| $\lambda_1 = 0$ | $\lambda_1' = 0$ | $\lambda_1'' = 0$ | $\lambda_1''' = 0$ |
| * | $\lambda_{2,3}' = \alpha^2 c_+/10 - \alpha a_+/2$ $\pm i\left(c_-\sqrt{\beta} + a_-\sqrt{\alpha}\right)/2$ | $\lambda_{2,3}'' = -\alpha^2 a_+/10 + \alpha c_+/2$ $\pm i\left(c_-\sqrt{\beta} - a_-\sqrt{\alpha}\right)/2$ | $\lambda_{2,3}''' = \lambda_{2,3}'' - e_+$ |
| $\lambda_{2,3} = (\beta c_+ - \alpha a_+)/2$ $\pm i\left(c_-\sqrt{\alpha} - a_-\sqrt{\beta}\right)/2$ | $\lambda_{4,5}' = \lambda_{2,3}$ | $\lambda_{4,5}'' = \lambda_{2,3}$ | $\lambda_{4,5}''' = \lambda_{4,5}'' = \lambda_{2,3}$ |
| * | $\lambda_{6,7}' = \beta^2 c_+/10 - \beta a_+/2$ $\pm i\left(c_-\sqrt{\alpha} - a_-\sqrt{\beta}\right)/2$ | $\lambda_{6,7}'' = -\beta^2 a_+/10 + \beta c_+/2$ $\pm i\left(c_-\sqrt{\alpha} + a_-\sqrt{\beta}\right)/2$ | $\lambda_{6,7}''' = \lambda_{6,7}'' - e_+$ |
| $\lambda_{4,5} = (\alpha c_+ - \beta a_+)/2$ $\pm i\left(c_-\sqrt{\beta} + a_-\sqrt{\alpha}\right)/2$ | $\lambda_{8,9}' = \lambda_{4,5}$ | $\lambda_{8,9}'' = \lambda_{4,5}$ | $\lambda_{8,9}''' = \lambda_{8,9}'' = \lambda_{4,5}$ |
| * | $\lambda_{10}' = 2c_+$ | $\lambda_{10}'' = -2a_+$ | $\lambda_{10}''' = \lambda_{10}'' - e_+$ |

The solutions of the above matrix equation reveal how the behaviors of the five agents evolve. Figure 3 shows the results of three typical solutions, representing the three states of the Yin-Yang network with the type-I interaction.

The following three initial conditions are used in the numerical solutions to generate the three states in Fig. 3:

$$\begin{bmatrix} (x_A(0), y_A(0)) \\ (x_B(0), y_B(0)) \\ (x_C(0), y_C(0)) \\ (x_D(0), y_D(0)) \\ (x_E(0), y_E(0)) \end{bmatrix} = \begin{bmatrix} (3.1,\ 4) \\ (1,\ 2) \\ (0,\ 0) \\ (-1,-2) \\ (-3,-4) \end{bmatrix}, \begin{bmatrix} (3,\ 4.1) \\ (1,\ 2) \\ (0,\ 0) \\ (-1,-2) \\ (-3,-4) \end{bmatrix}, \begin{bmatrix} (3,\ 4) \\ (1,\ 2) \\ (0,\ 0) \\ (-1,-2) \\ (-3,-4) \end{bmatrix}. \tag{14}$$

We note that these three sets of initial conditions only have slight differences in $x_A(0)$ and $x_B(0)$. The constant rates of the four effects GA, GR, OA, and OR are chosen as $a = 2, b = 3, c = 0.5$, and $d = 1.5$ in the numerical solutions. The time responses obtained from the first set of initial conditions show that the five Yin components $\begin{bmatrix} x_A & x_B & x_C & x_D & x_E \end{bmatrix}^T$

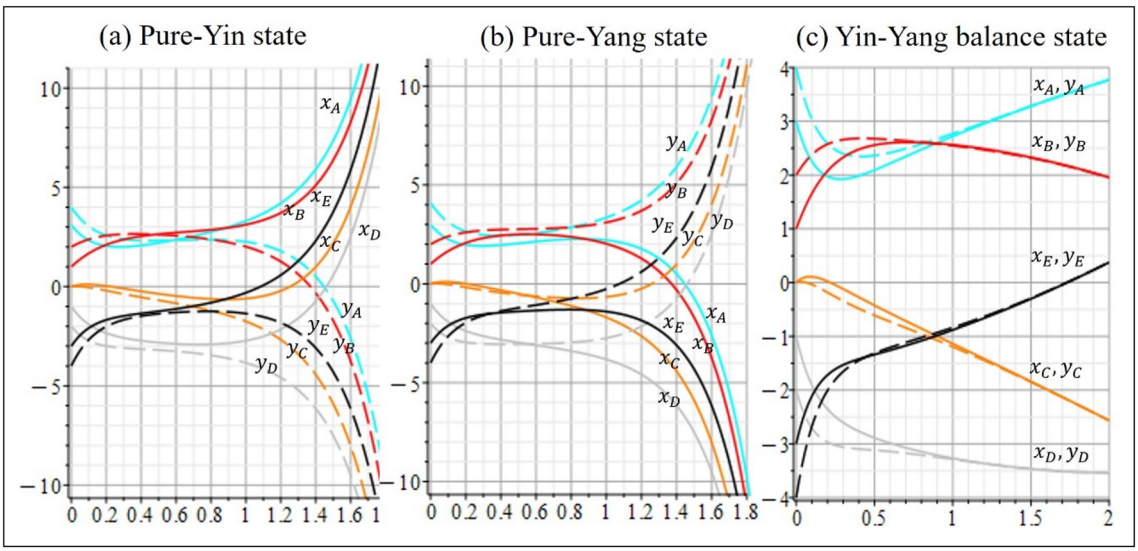



**Fig. 3** Three typical states of the Yin-Yang network with type-I interaction. **a** The five Yin components increase to merge together and dominate over the Yang components, forming the pure Yin state. **b** The five Yang components increase to merge together and dominate over the Yin components, forming the pure Yang state. **c** Individual agents' Yin and Yang components first merge to form a Yin-Yang balance state and then diverge to infinity



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increase with time and gradually merge into one curve (solid line), while the five Yang components $\begin{bmatrix} y_A & y_B & y_C & y_D & y_E \end{bmatrix}^T$ decrease with time and merge into another curve (dashed line). The second set of initial conditions shows a similar trend of Yin-Yang separation, but the difference is that the five Yang components become dominant. The result of the third set of initial conditions completely differs from the previous two. In Fig. 3c, the Yin and Yang components of the same agent merge into one curve and then diverge to infinity. In this case, the Yin and Yang components are balanced in the same agent but are unbalanced between different agents.

The three states observed in Fig. 3 are inherent characteristics of the Yin-Yang network with the type-I interaction, and they reflect the eigenstructure of the network $\mathbf{A}\prime$ given by Eq. (13). The phenomenon of Yin-Yang separation during the divergent process can be explained by the dominant eigenvalue $\lambda_{10}\prime = 2c_+ > 0$ and its related eigenvector $\mathbf{v}_{10}\prime$. The eigenvector $\mathbf{v}_{10}\prime$ satisfies the following characteristic equation:

$$\left(\mathbf{A}' - \lambda_{10}\prime\mathbf{I}\right)\mathbf{v}_{10}\prime = \begin{bmatrix} \Delta & 0 & b & 0 & 0 & -d & 0 & -c & a & 0 \\ 0 & \Delta & 0 & b & -d & 0 & -c & 0 & 0 & a \\ a & 0 & \Delta & 0 & b & 0 & 0 & -d & 0 & -c \\ 0 & a & 0 & \Delta & 0 & b & -d & 0 & -c & 0 \\ 0 & -c & a & 0 & \Delta & 0 & b & 0 & 0 & -d \\ -c & 0 & 0 & a & 0 & \Delta & 0 & b & -d & 0 \\ 0 & -d & 0 & -c & a & 0 & \Delta & 0 & b & 0 \\ -d & 0 & -c & 0 & 0 & a & 0 & \Delta & 0 & b \\ b & 0 & 0 & -d & 0 & -c & a & 0 & \Delta & 0 \\ 0 & b & -d & 0 & -c & 0 & 0 & a & 0 & \Delta \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = 0, \tag{15}$$

with $\Delta = \delta - 2c_+ = -(a_+ + c_+)$. We notice that $\mathbf{v}_{10}\prime$ has the following special Yin-Yang separation structure:

$$\mathbf{v}_{10}\prime = \begin{bmatrix} 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix}^T, \tag{16}$$

where the five Yin components $\begin{bmatrix} x_A & x_B & x_C & x_D & x_E \end{bmatrix}^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \end{bmatrix}^T$ are all equal to 1, while the five Yang components $\begin{bmatrix} y_A & y_B & y_C & y_D & y_E \end{bmatrix}^T = \begin{bmatrix} -1 & -1 & -1 & -1 & -1 \end{bmatrix}^T$ are all equal to −1. The form of eigenvector $\mathbf{v}_{10}\prime$ represents the mode shape associated with $\lambda_{10}\prime$. Since $\lambda_{10}\prime$ is the dominant eigenvalue, the related eigenvector $\mathbf{v}_{10}\prime$ governs the asymptotical behavior of the state vector $\mathbf{X}(t)$ as $t \to \infty$, which can be expressed analytically as

$$\mathbf{X}(t) \approx c_{10}\mathbf{v}_{10}\prime e^{\lambda_{10}\prime t} = \frac{1}{10}\left(\sum\nolimits_{i=A}^{E} x_i(0) - \sum\nolimits_{i=A}^{E} y_i(0)\right)\mathbf{v}_{10}\prime e^{\lambda_{10}\prime t}, \tag{17}$$

where the constant $c_{10} = (1/10)(\sum_i x_i(0) - \sum_i y_i(0))$ is determined by the initial conditions. Among the three sets of initial conditions in Eq. (14), the first set makes $c_{10} > 0$, while the second set makes $c_{10} < 0$, thus producing the results of Yin-Yang separation with opposite trends in Fig. 3a and b. The third set of initial conditions just makes $c_{10} = 0$, causing state vector $\mathbf{X}(t)$ independent of $\lambda_{10}\prime$ and $\mathbf{v}_{10}\prime$. For this special case, the behavior of X(t) is dominated by the second largest eigenvalue $\lambda_4\prime$, and Eq. (17) is replaced by $\mathbf{X}(t) \approx c_4\mathbf{v}_4\prime e^{\lambda_4\prime t}$. The divergent speed of the time response in Fig. 3c reflects the real part of $\lambda_4\prime$, while the trend of the Yin-Yang combination of each agent reflects the structure of the eigenvector $\mathbf{v}_4\prime$.

The Yin-Yang network produced by the type-I interaction causes the divergence of the network and separates the five Yin components from the five Yang components. The network eventually tends to a pure Yang state or a pure Yin state, which is the realm pursued by traditional Taoist practitioners but runs counter to the Yin-Yang balance that TCM aims to achieve. In order to achieve a balanced state of Yin and Yang, we must consider the type-II interaction.

## 4 Type-II Interaction: Yin-Yang cooperation and Yin-Yin competition

The type-II interaction is in line with Coulomb's law that unlike charges attract each other and like charges repel each other. Figures 2c and d take agent A as an example to illustrate the action and reaction received by agent A's Yin component (real part, $x_A$) and Yang component (imaginary part, $y_A$) from the other four agents. According to the type-II interaction, mutual generation occurs between a Yin component and a Yang component, while mutual competition



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occurs between two Yin or two Yang Components. Adding the four actions and reactions exerting on $x_A$ in Fig. 2c, we get the change rate $\dot{x}_A$. In the same way, by adding the four actions and reactions exerting on $y_A$ in Fig. 2d, we get the change rate $\dot{y}_A$. The resulting equations for the ten components can be put into the following matrix form:

$$\begin{bmatrix} \dot{x}_A \\ \dot{y}_A \\ \dot{x}_B \\ \dot{y}_B \\ \dot{x}_C \\ \dot{y}_C \\ \dot{x}_D \\ \dot{y}_D \\ \dot{x}_E \\ \dot{y}_E \end{bmatrix} = \begin{bmatrix} \delta & 0 & 0 & b & -d & 0 & -c & 0 & 0 & a \\ 0 & \delta & b & 0 & 0 & -d & 0 & -c & a & 0 \\ 0 & a & \delta & 0 & 0 & b & -d & 0 & -c & 0 \\ a & 0 & 0 & \delta & b & 0 & 0 & -d & 0 & -c \\ -c & 0 & 0 & a & \delta & 0 & 0 & b & -d & 0 \\ 0 & -c & a & 0 & 0 & \delta & b & 0 & 0 & -d \\ -d & 0 & -c & 0 & 0 & a & \delta & 0 & 0 & b \\ 0 & -d & 0 & -c & a & 0 & 0 & \delta & b & 0 \\ 0 & b & -d & 0 & -c & 0 & 0 & a & \delta & 0 \\ b & 0 & 0 & -d & 0 & -c & a & 0 & 0 & \delta \end{bmatrix} \begin{bmatrix} x_A \\ y_A \\ x_B \\ y_B \\ x_C \\ y_C \\ x_D \\ y_D \\ x_E \\ y_E \end{bmatrix} \rightarrow \dot{\mathbf{X}}'' = \mathbf{A}''\mathbf{X}''. \tag{18}$$

The third column of Table 1 lists the ten eigenvalues of the system matrix $\mathbf{A''}$ in decreasing order of their real parts. Unlike the system matrix $\mathbf{A}'$, which is always unstable, the stability of the system matrix $\mathbf{A''}$ depends on the strength ratio between the cooperative and the competitive interactions.

### 4.1 Stability of TCM

To guarantee the stability of Eq. (18), the maximum real part of the nonzero eigenvalues must be negative, that is

$$\sigma''_{2,3} < 0 \rightarrow \frac{a_+}{c_+} > \frac{5}{\alpha} = \frac{5+\sqrt{5}}{2} = \varphi^2 + 1, \tag{19}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The inequality (19) reveals that the ratio of the generating interaction rate $a_+ = a + b$ to the overcoming interaction rate $c_+ = c + d$ must be greater than $\varphi^2 + 1$to ensure the stability of the Yin-Yang network with the type-II interaction.

When condition (19) is satisfied, the dominant eigenvalue of $\mathbf{A''}$ is $\lambda''_1 = 0$ and its related eigenvector is $\mathbf{v}''_1 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}^T$ with elements all equal to one. In terms of $\mathbf{v}''_1$, the steady-state solution of $\mathbf{X}''(t)$ can be expressed as

$$\mathbf{X}''_{ss} = \lim_{t\to\infty} \mathbf{X}''(t) = c''_1 \mathbf{v}''_1 e^{\lambda''_1 t} = c''_1 \mathbf{v}''_1. \tag{20}$$

Because all the elements in the vector $\mathbf{v}''_1$ are equal to 1, the ten components in $\mathbf{X}''(t)$ all converge to $c''_1$, which is the consensus of the Yin-Yang network and can be determined by the initial conditions as

$$c''_1 = \frac{1}{10}\left(\sum_{i=A}^{E} x_i(0) + \sum_{i=A}^{E} y_i(0)\right). \tag{21}$$

The consensus achieved by $\mathbf{X}''_{ss}$ has two meanings of balance: the Yin-Yang balance in individual agent and the Yin-Yang balance between different agents.

Two sets of network parameters are used in the computation for comparison:

$$\text{(A)}a = 4,\ b = 6,\ c = 0.5,\ d = 1.5 : (a+b)/(c+d) = 5 > \varphi^2 + 1$$

$$\text{(B)}a = 2,\ b = 4,\ c = 0.5,\ d = 1.5 : (a+b)/(c+d) = 3 < \varphi^2 + 1$$

Set A meets the stability condition (19) and is expected to establish a balanced network. The rate ratio used in set B is less than $\varphi^2 + 1$, which will lead to an unstable network. The solutions to Eq. (18) subjected to two sets of parameters are graphically illustrated in Fig. 4. Figure 4a show the time responses of the five agents by using the parameters in set A, indicating that the five agents with their Yin and Yang components all converge to the same equilibrium value. Figure 4b



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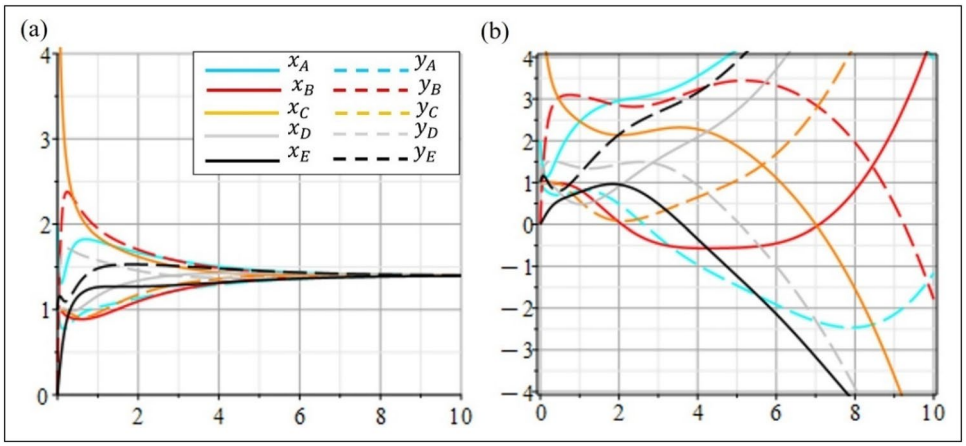



**Fig. 4** The time responses of the complex five-agent network with type-II interaction. **a** The time responses of the five agents converge to the same equilibrium value by using the parameters in set A, which satisfies the stability condition (19). **b** The time responses diverge to infinity by using the parameters in set B, which violates the stability condition (19). The numerical solutions are obtained by solving Eq. (18) with initial conditions: $(x_A(0), y_A(0)) = (2, 1), (x_B(0), y_B(0)) = (1, 0), (x_C(0), y_C(0)) = (6, 1), (x_D(0), y_D(0)) = (1, 1)$, and $(x_E(0), y_E(0)) = (0, 1)$

show the time responses by using the parameters in set B, which violate the stability condition and yield unstable network with all the Yin-Yang components diverging to infinity.

Although networks $\mathbf{A}'$ and $\mathbf{A}''$ have different stabilities, they have common characteristics. If we let $y_i(t) = x_i(t)$, $i = A, B, C, D, E$, in Eq. (13) and Eq. (18), they both reduce to the same form

$$\begin{bmatrix} \dot{x}_A \\ \dot{x}_B \\ \dot{x}_C \\ \dot{x}_D \\ \dot{x}_E \end{bmatrix} = \begin{bmatrix} \sigma & b & -d & -c & a \\ a & \sigma & b & -d & -c \\ -c & a & \sigma & b & -d \\ -d & -c & a & \sigma & b \\ b & -d & -c & a & \sigma \end{bmatrix} \begin{bmatrix} x_A \\ x_B \\ x_C \\ x_D \\ x_E \end{bmatrix} \rightarrow \dot{\mathbf{X}} = \mathbf{A}\mathbf{X}. \tag{22}$$

This is just the ordinary five-agent network shown in Fig. 1b, where the Yin and Yang components of each agent are balanced and there is no need to distinguish between them. Since the ordinary five-agent network $\mathbf{A}$ is a degenerate case of the complex five-agent networks $\mathbf{A}'$ and $\mathbf{A}''$, the eigenvalues of $\mathbf{A}$ are retained among the eigenvalues of $\mathbf{A}'$ and $\mathbf{A}''$, as can be seen from Table 1 (highlighted in red).

The stability of the ordinary five-agent network $\mathbf{A}$ given by Eq. (22) requires that the real part of its dominant eigenvalue $\lambda_{23}$ must be negative. From Table 1, the condition $\mathrm{Re}(\lambda_{23}) < 0$ can be expressed as

$$\frac{a_+}{c_+} > \frac{3 + \sqrt{5}}{2} = \varphi^2. \tag{23}$$

This is the main result reported in Ref. [7]. In comparison with Eq. (19), it appears that that when the five-agent network takes the Yin-Yang duality into account, the ratio $a_+/c_+$ must increase by 1 to ensure the stability of the network. This is because the Yin-Yang duality introduces an additional imbalanced state to the network, so greater cooperative interactions (larger $a_+$) are needed to ensure the stability of the overall network.

### 4.2 Internal feedback structure of TCM

The ability of the complex-valued five-agent network to recover from the initial imbalance to the equilibrium state as observed in Fig. 4a is attributed to its inherent feedback mechanism. There are two feedback signals in the five-agent network. For the generating action, its feedback signal originates from the generating reaction, both of which occur at the same time, but in opposite directions; while for the overcoming action, its feedback signal originates from the overcoming reaction,



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both of which also occur at the same time, but in opposite directions. According to the two feedback signals, Eq. (18) can be rewritten as a state-feedback form as

$$\dot{\mathbf{X}}'' = \mathbf{A}''\mathbf{X}'' = \mathbf{A}''_{\text{open}}\mathbf{X}'' + \mathbf{u}_b + \mathbf{u}_d = \mathbf{A}''_{\text{open}}\mathbf{X}'' - \mathbf{K}_b\mathbf{X}'' + \mathbf{K}_d\mathbf{X}'', \tag{24}$$

where $\mathbf{A}''_{\text{open}}$ is the open-loop system matrix containing the generating action and overcoming action only, $\mathbf{K}_b$ is the negative-feedback gain matrix containing the generating reaction, and $\mathbf{K}_d$ is the positive-feedback gain matrix containing the overcoming reaction, respectively defined as

$$\mathbf{A}''_{\text{open}} = \begin{bmatrix} \delta_{ca} & 0 & 0 & 0 & 0 & 0 & -c & 0 & 0 & a \\ 0 & \delta_{ca} & 0 & 0 & 0 & 0 & 0 & -c & a & 0 \\ 0 & a & \delta_{ca} & 0 & 0 & 0 & 0 & 0 & -c & 0 \\ a & 0 & 0 & \delta_{ca} & 0 & 0 & 0 & 0 & 0 & -c \\ -c & 0 & 0 & a & \delta_{ca} & 0 & 0 & 0 & 0 & 0 \\ 0 & -c & a & 0 & 0 & \delta_{ca} & 0 & 0 & 0 & 0 \\ -d & 0 & -c & 0 & 0 & a & \delta_{ca} & 0 & 0 & b \\ 0 & -d & 0 & -c & a & 0 & 0 & \delta_{ca} & b & 0 \\ 0 & b & -d & 0 & -c & 0 & 0 & a & \delta_{ca} & 0 \\ b & 0 & 0 & -d & 0 & -c & a & 0 & 0 & \delta_{ca} \end{bmatrix},$$

$$\mathbf{K}_b = b\begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \mathbf{K}_d = d\begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

where $\delta_{ca} = c - a$. The negative feedback signal $\mathbf{u}_b = -\mathbf{K}_b\mathbf{X}''$ helps to reduce the gap between agents, while the positive feedback signal $\mathbf{u}_d = \mathbf{K}_d\mathbf{X}''$ tends to enlarge the gap between agents. Under the simultaneous action of these two feedback signals, whether the human body can maintain a balanced state is determined by stability condition (19), which is a measure of the relative magnitude between the negative and positive feedback signals. If Eq. (19) is not satisfied as the case considered in Fig. 4b, the five internal organs of the human body lose their self-balancing function, and the human body is in a state of illness requiring medical treatment. Based on the established mathematical model, the role of medical treatment is equivalent to inputting an external control signal $\mathbf{u}_{\text{ext}}$ to Eq. (24) in addition to the existing internal feedback signals $\mathbf{u}_b$ and $\mathbf{u}_d$, so that the controlled system can be restored to its original stable equilibrium state.

### 4.3 The interaction of the human body with the environment

The model expressed by Eq. (24) is only valid for closed systems whose total energy is conserved; however, the human body is not a closed system. In order to maintain physiological functions, the human body must input energy from the environment and dissipate energy to the environment, so the equilibrium state achieved is actually a dynamic state, not a stationary state. To account for the energy interaction between the human body and the external environment, an environmental input $\mathbf{w}(t)$ is added to Eq. (24) to complete the model of TCM:

$$\dot{\mathbf{X}}'' = \mathbf{A}''\mathbf{X}'' + \mathbf{w}(t) + \mathbf{u}_{\text{ext}}\left(\mathbf{X}''\right), \tag{25}$$



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where $\mathbf{u}_{\text{ext}}(\mathbf{X}'')$ is the aforementioned medical treatment dependent on the current state $\mathbf{X}''$ of the human body. Here we do not intend to establish a precise mathematical form for $\mathbf{w}(t)$, but to analyze its impact on the internal feedback mechanism of the human body.

To deal with the expression for $\mathbf{w}(t)$, we recall that the agents of the Wuxing network in the human body are the five organs of liver (agent A), heart (agent B), spleen (agent C), lung (agent D), and kidney (agent E). The amount of nutrients they store is represented by $x_i$, and the energy stored is represented by $y_i$. The main energy consumption in the human body is mainly used to maintain the operation of the heart, and its quantitative index is $y_B$, which represents the energy stored in the heart. Since the heart continuously consumes energy, if there is no energy input from the environment, the value of $y_B$ decreases over time, that is, the rate of change $\dot{y}_B$ is negative. On the other hand, the main energy input to the human body comes from nutrients produced after digestion of food, and the representative index is $x_C$, which denotes the nutrients stored in the spleen. Since the human body digests food periodically, the value of $x_C$ changes periodically over time.

Considering the above energy transfer between the human body and the environment, a simplified dimensionless form of $\mathbf{w}(t)$ can be expressed as follows.

$$\mathbf{w}(t) = \left[\,0\ 0\ 0\ -1\ P(t)\ 0\ 0\ 0\ 0\ 0\,\right]^T. \tag{26}$$

The fourth element of $\mathbf{w}(t)$ represents the input signal $\dot{y}_B = -1$, indicating that the heart consumes one unit of energy per unit time. The fifth element $P(t)$ of $\mathbf{w}(t)$ represents the nutrients stored in the spleen, which is a periodic time function as shown in Fig. 5b. The period of the pulse signal $P(t)$ in the figure is ten units of time. The pulse height in the first two units of time is five, and is zero elsewhere, which schematically expresses the periodic operation of the digestive organs.

Substituting $\mathbf{w}(t)$ from Eq. (26) into Eq. (25) and adopting the same parameter setting as used in Fig. 4a to solve for $\mathbf{X}''(t)$, we obtain the numerical results as shown in Fig. 5. In this numerical simulation we want to test whether the internal feedback mechanism of the human body can maintain the balance between the five organs under the action of the input signal $\mathbf{w}(t)$ but without the medical treatment $\mathbf{u}_{\text{ext}}$. In Fig. 5a we only consider the energy consumption of the heart ($\dot{y}_B = -1$), and ignore the energy input from the environment ($P(t) = 0$). The results show that the ten components of $\mathbf{X}''(t)$ cannot converge to the equilibrium state, but all decrease following the environmental input $\mathbf{w}(t)$. According to the input signal $\dot{y}_B = -1$, the heart consumes thirty units of energy in thirty units of time. Without the energy input from the environment, these thirty units of consumed energy are evenly compensated by the ten components of $\mathbf{X}''(t)$ through the cooperative and competitive interactions between the five organs. Therefore, each component reduces energy by about three units in thirty units of time. This is exactly the energy consumption rate we observed in Fig. 5a.

Figure 5b shows that when there is energy input $P(t)$ from the environment, the ten components of $\mathbf{X}''(t)$ still have a periodic downward trend. However, because the energy gained from the human body's periodic digestion of food compensates for the energy consumption of cardiac activity, the average energy of the human body does not drop

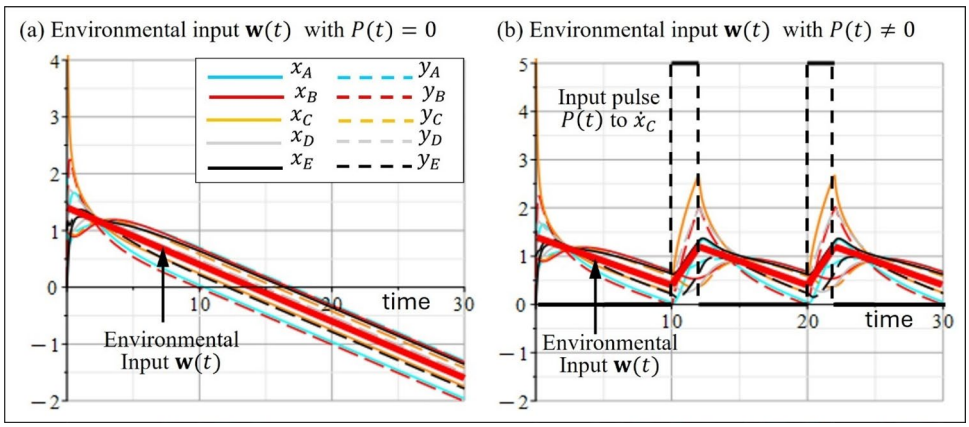



**Fig. 5** The responses of the complex five-agent network to the environmental input $\mathbf{w}(t)$. **a** The response to $\mathbf{w}(t)$ with $P(t) = 0$ shows that all the ten components of $\mathbf{X}''(t)$ decrease continuously along a fixed slope due to the constant energy consumption of the heart. **b** The response to $\mathbf{w}(t)$ with $P(t) \neq 0$ shows that the energy consumption of cardiac activity is compensated by the energy gained from the human body's periodic digestion of food such that the total energy changes in a zigzag pattern (thick red line)



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continuously but shows a zigzag change. According to the pulse wave pattern of $P(t)$, the energy input to the human body in every ten units of time is approximately ten units of energy, which is equivalent to an average input rate of one unit energy per unit time. Such an energy input rate just balances the energy consumption of cardiac activity, thus presenting the result that the average energy changes in a zigzag pattern as shown in Fig. 5b.

As can be observed from the numerical simulation in Fig. 5, the human body's internal feedback mechanism has strong adaptability to the environmental change. When there is no energy transfer with the environment, the human body's total energy is conservative, and the internal feedback mechanism automatically makes the human body converge to an equilibrium state that does not change with time (see Fig. 4a). When there is energy transfer, the internal feedback mechanism causes the total energy of the human body to follow closely the environmental input command $\mathbf{w}(t)$, such as the negative ramp command in Fig. 5a and the zigzag wave command in Fig. 5b.

### 4.4 Yin-Yang diagram (Tai Chi symbol) under critical stability

Yin-Yang diagram, also called Tai Chi diagram, is a totem of the Yin-Yang complementarity and was ever used by Niels Bohr to symbolize the wave-particle duality. Since we model the Yin-Yang complementarity by the real-imaginary complementarity, the changes in each agent's real and imaginary components are expected to form a Yin-Yang diagram. In Fig. 1c and d, each agent has a corresponding Yin-Yang diagram, which is not drawn casually but is based on the time evolution of the real component $x(t)$ and imaginary component $y(t)$ of each agent.

From Table 1, the dominant eigenvalue of the complex five-agent network $\mathbf{A}''$ with type-II interaction is

$$\lambda''_{2,3} = \sigma''_{2,3} \pm i\omega''_{2,3} = -\frac{\alpha^2 a_+}{10} + \frac{\alpha c_+}{2} \pm \frac{\mathrm{i}}{2}\left(a_-\sqrt{\alpha} - c_-\sqrt{\beta}\right), \tag{27}$$

where the real part $\sigma''_{2,3}$ determines the stability of the network and the imaginary part $\omega''_{2,3}$ determines the vibration frequency of the five-agent network. As $\sigma''_{2,3} < 0$, the ten components of $\mathbf{X}''(t)$ converge to their consensus, as shown in Fig. 4a. In contrast, as $\sigma''_{2,3} > 0$, the ten components of $\mathbf{X}''(t)$ all diverge to infinity, as shown in Fig. 4b. When $\sigma''_{2,3} = 0$, the ten components of $\mathbf{X}''(t)$ neither converge nor diverge, but maintain a constant-amplitude vibration, corresponding to the case of critical stability. Let $\sigma''_{2,3} = 0$, we obtain the condition for critical stability as

$$\frac{a_+}{c_+} = \frac{5}{\alpha} = \beta. \tag{28}$$

Under this condition, the five-agent network exhibits sinusoidal oscillation with frequency given by

$$\omega''_{2,3} = \left(a_-\sqrt{\alpha} - c_-\sqrt{\beta}\right)/2. \tag{29}$$

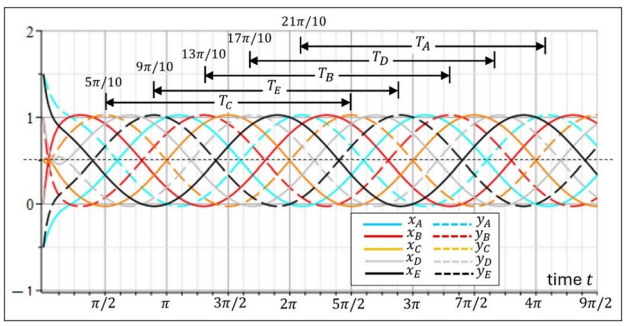



**Fig. 6** Time responses of the complex five-agent network under the critical stability. After a short transient response, the ten network components sequentially exhibit sinusoidal oscillations with fixed amplitude and frequency. The five intervals represent the oscillation periods of the five agents, within which the real part $x(t)$ and the imaginary part $y(t)$ of each agent are read to construct its Yin-Yang diagram. The data are obtained by solving Eq. (18) with initial conditions: $(x_A(0), y_A(0)) = (-0.5, 1.5)$, $(x_B(0), y_B(0)) = (0,1)$, $(x_C(0), y_C(0)) = (0.5, 0.5)$, $(x_D(0), y_D(0)) = (1,0)$, and $(x_E(0), y_E(0)) = (1.5, -0.5)$



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Figure 6 shows the time response of the Yin-Yang five-agent network under critical stability. The parameters used are $c = d = 1$, and the vibration frequency under critical stability is set to be $\omega''_{2,3} = 1$, that is, the corresponding vibration period is 2π. The related parameters $a$ and $b$ can be determined from Eq. (28) and Eq. (29) as $a = \beta + 1/\sqrt{\alpha}$ and $b = \beta - 1/\sqrt{\alpha}$. Substituting the above parameter settings into Eq. (18) for numerical solution, we obtain the time response as depicted in Fig. 6. After a short transient response, the ten network components sequentially exhibit sinusoidal oscillations with fixed amplitude and fixed frequency. The five intervals shown in Fig. 6 respectively denote the vibration periods of the five agents, all equal to $2\pi$. The data within the periods will be used to construct the Yin-Yang diagram corresponding to each agent.

The Yin-Yang diagram can be drawn for any periodic system composed of a complementary pair called Yin-Yang pair, whose quantitative measures $x$ and $y$ are normalized between 0 and 1. The growth and decline of the Yin-Yang pair are conveyed through the interlacing changes of $x$ (in black) and $y$ (in white), which in turn forms the Yin-Yang diagram. Figure 7a takes agent C as an example to explain how to construct its Yin-Yang diagram. In terms of the quantitative measures $x_C(t)$ and $y_C(t)$ computed from Eq. (18), the inner curve of the Yin-Yang diagram for agent C is defined by the following polar coordinates $(r, \theta)$

$$r = \begin{cases} y_C(\theta), & 0 \le \theta \le \pi \\ x_C(\theta), & \pi \le \theta \le 2\pi \end{cases} = \begin{cases} y_C(t - t_0), & t_0 \le t \le t_0 + \pi \\ x_C(t - t_0), & t_0 + \pi \le t \le t_0 + 2\pi \end{cases}, \tag{30}$$

and its outer curve is defined by $r = x_C(\theta) + y_C(\theta)$. The azimuth coordinate $\theta$ is equal to $t - t_0$ starting from the positive vertical axis and increasing clockwise.

As can be seen from Fig. 7b, agent C's Yin-Yang diagram starts at $t_0 = \pi/2$ and ends at $t_4 = 5\pi/2$, which exactly constitutes a period. According to the sinusoidal solutions of $x_C(t)$ and $y_C(t)$ within the period, the polar Eq. (30) can be expressed analytically as

$$r = \begin{cases} (1 + \cos\theta)/2, & 0 \le \theta \le \pi \\ (1 - \cos\theta)/2, & \pi \le \theta \le 2\pi \end{cases}, \tag{31}$$

which is composed of two sections of cardioid. The property $r = x_C(\theta) + y_C(\theta) = 1$ indicates that the periphery of the Yin-Yang diagram is a unit circle. Surprisingly, the Yin-Yang diagram described by Eq. (31) is the same as that obtained by another completely unrelated system, which involves a particle's quantum tunneling motion via a step barrier [6]. This result shows that the Yin-Yang diagram is a universal existence and independent of the system's physical characteristics [26]. The Yin-Yang diagrams of the other four agents can be constructed in the same manner by their time responses in Fig. 6. The appearance of the five Yin-Yang diagrams is found to be the same, except that their starting time $t_0$ is different due to different phase shifts, as shown in Fig. 6.

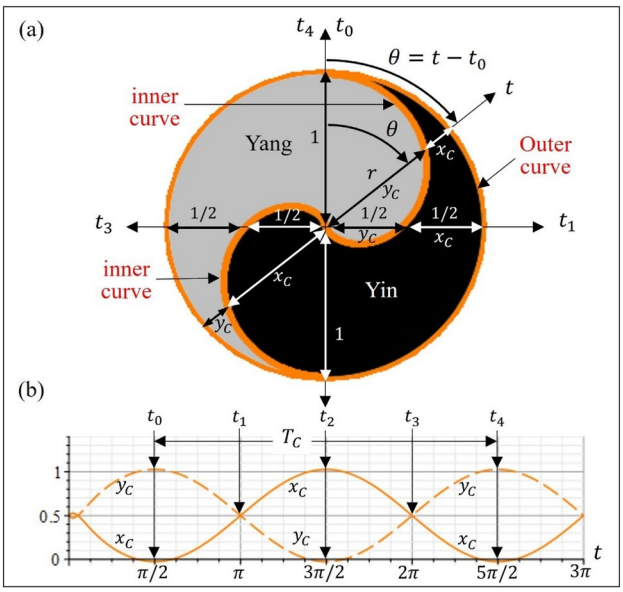



**Fig. 7** Drawing the Yin-Yang diagram for agent C from the time response of $\boldsymbol{x_C(t)}$ **and** $\boldsymbol{y_C(t)}$**. a** The outer curve of the Yin-Yang diagram is defined by $r = y_C(t) + x_C(t)$, and the inner curve is defined by $r = y_C(t)$ for the right-half diagram and $r = x_C(t)$ for the left-half diagram. **b** The values of $y_C(t)$ and $x_C(t)$ are solved from Eq. (18) to construct the agent C's Yin-Yang diagram according to Eq. (34)



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## 5 The complex five-agent network with self-generation

The preceding discussion ignores the internal dynamics of agents, which amounts to assuming that there is no dynamic relationship between the real and imaginary components of the same agent. The five agents' mutual generation and competition belong to the external dynamics between agents, while self-generation is the internal dynamics, which refers to the mechanism by which the Yin-Yang components of the same agent transform into each other to promote their balance. From the viewpoint of TCM, self-generation considers the mutual conversion in the same organ between the tangible nutrients and the intangible energy released after the nutrients are oxidized.

Taking agent A as an example, the mutual conversion between its Yin component $x_A$ and Yang component $y_A$ can be expressed by the following relationship:

$$\dot{x}_A = e\left(y_A - x_A\right), \tag{32}$$

$$\dot{y}_A = f\left(x_A - y_A\right), \tag{33}$$

where $e > 0$ and $f > 0$ represent the bidirectional self-generation rates. When $y_A > x_A$, Eq. (32) gives $\dot{x}_A > 0$, which means that when $y_A$ is more than $x_A$, $y_A$ is converted into $x_A$, causing the value of $x_A$ to increase; at the same time, Eq. (33) gives the decreasing rate of $y_A$. On the contrary, if $x_A > y_A$, $x_A$ is converted to $y_A$, so that the value of $y_A$ increases and the value of $x_A$ decreases.

The self-generation effects of other agents can be expressed in a way similar to Eq. (32) and Eq. (33). Adding the self-generation relation of each agent to Eq. (18), we obtain the governing equations for the complex five-agent network with self-generation as

$$\begin{bmatrix} \dot{x}_A \\ \dot{y}_A \\ \dot{x}_B \\ \dot{y}_B \\ \dot{x}_C \\ \dot{y}_C \\ \dot{x}_D \\ \dot{y}_D \\ \dot{x}_E \\ \dot{y}_E \end{bmatrix} = \begin{bmatrix} \delta & e & 0 & b & -d & 0 & -c & 0 & 0 & a \\ f & \delta\prime & b & 0 & 0 & -d & 0 & -c & a & 0 \\ 0 & a & \delta & e & 0 & b & -d & 0 & -c & 0 \\ a & 0 & f & \delta\prime & b & 0 & 0 & -d & 0 & -c \\ -c & 0 & 0 & a & \delta & e & 0 & b & -d & 0 \\ 0 & -c & a & 0 & f & \delta\prime & b & 0 & 0 & -d \\ -d & 0 & -c & 0 & 0 & a & \delta & e & 0 & b \\ 0 & -d & 0 & -c & a & 0 & f & \delta\prime & b & 0 \\ 0 & b & -d & 0 & -c & 0 & 0 & a & \delta & e \\ b & 0 & 0 & -d & 0 & -c & a & 0 & f & \delta\prime \end{bmatrix} \begin{bmatrix} x_A \\ y_A \\ x_B \\ y_B \\ x_C \\ y_C \\ x_D \\ y_D \\ x_E \\ y_E \end{bmatrix} \rightarrow \dot{\mathbf{X}}''' = \mathbf{A}'''\mathbf{X}''', \tag{34}$$

where $\delta = -\left(a_+ + e - c_+\right)$ and $\delta\prime = -\left(a_+ + f - c_+\right)$.

Up to this stage, we have established four system matrices: system matrix $\mathbf{A}$ for the ordinary five-agent network given by Eq. (22), system matrix $\mathbf{A}'$ for the complex five-agent networks with the type I interaction given by Eq. (13), system matrix $\mathbf{A}''$ for the complex five-agent networks with the type II interaction given by Eq. (18), and system matrix $\mathbf{A}'''$ for the complex five-agent network with self-generation given by Eq. (34). If we equal the real components $x_i$ to the imaginary components $y_i$ in the complex networks, all the three complex networks automatically reduce to the real ordinary network, that is, $\mathbf{A}\prime = \mathbf{A}\prime\prime = \mathbf{A}\prime\prime\prime = \mathbf{A}$. Therefore, the ordinary network $\mathbf{A}$ forms the basic structure of the complex networks, and its eigenvalues are the common eigenvalue of the four networks as highlighted in red in Table 1.

The relative positions of the eigenvalues of the three networks $\mathbf{A}$, $\mathbf{A}''$, and $\mathbf{A}'''$ on the real axis are shown in Fig. 8. The red circle denotes the common eigenvalue of the three networks, the blue triangle denotes the eigenvalue unique to $\mathbf{A}''$, and the yellow square denotes the eigenvalue unique to $\mathbf{A}'''$. It can be seen that except for the common eigenvalues, the eigenvalues of $\mathbf{A}'$ and $\mathbf{A}'''$ have a fixed difference $e_+$. Meanwhile, we notice that except for the common eigenvalues, the eigenvalues of $\mathbf{A}'$ and $\mathbf{A}'''$ have a fixed difference $e_+ = e + f$. According to Fig. 8, as the self-generation rate $e_+ = e + f$ increases, the eigenvalues unique to $\mathbf{A}'''$ move to the left, causing the network $\mathbf{A}'''$ to converge faster. When $\sigma_{2,3}''' = \sigma_{2,3}'' - e_+$ decreases to a value smaller than $\sigma_{4,5}'''$, the dominant eigenvalue becomes $\sigma_{4,5}'''$ instead of $\sigma_{2,3}'''$ and the increase in the self-generation rate $e_+$ no longer affects the convergent speed of the network $\mathbf{A}'''$. Let $\sigma_{4,5}''' = \sigma_{2,3}'''$, the critical value of $e_+$ can be obtained as



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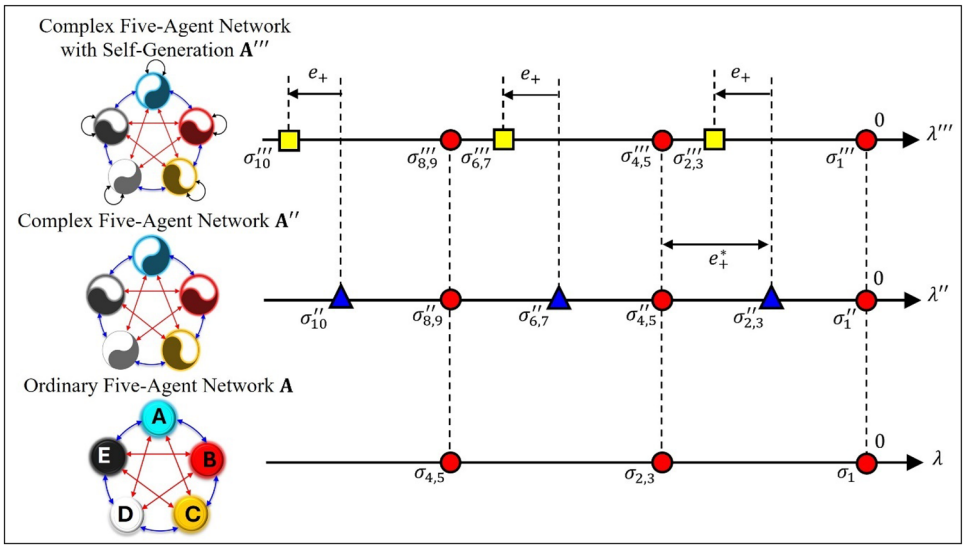



**Fig. 8** The eigenspectrum of the three types of five-agent network. $\lambda$, $\lambda''$, and $\lambda'''$ denote the eigenvalues of the networks $\mathbf{A}$, $\mathbf{A}''$, and $\mathbf{A}'''$, respectively. The red circle is the common eigenvalue of the three networks, the blue triangle is the eigenvalue unique to the complex five-agent network $\mathbf{A}''$, and the yellow square is the eigenvalue unique to the complex five-agent network $\mathbf{A}'''$ with self-conversion, which moves to the left as the self-generation rate $e_+$ increases. As $e_+$ increases beyond $e_+^*$, $\sigma_{4,5}'''$ becomes the dominant eigenvalues of $\mathbf{A}'''$ instead of $\sigma_{2,3}'''$

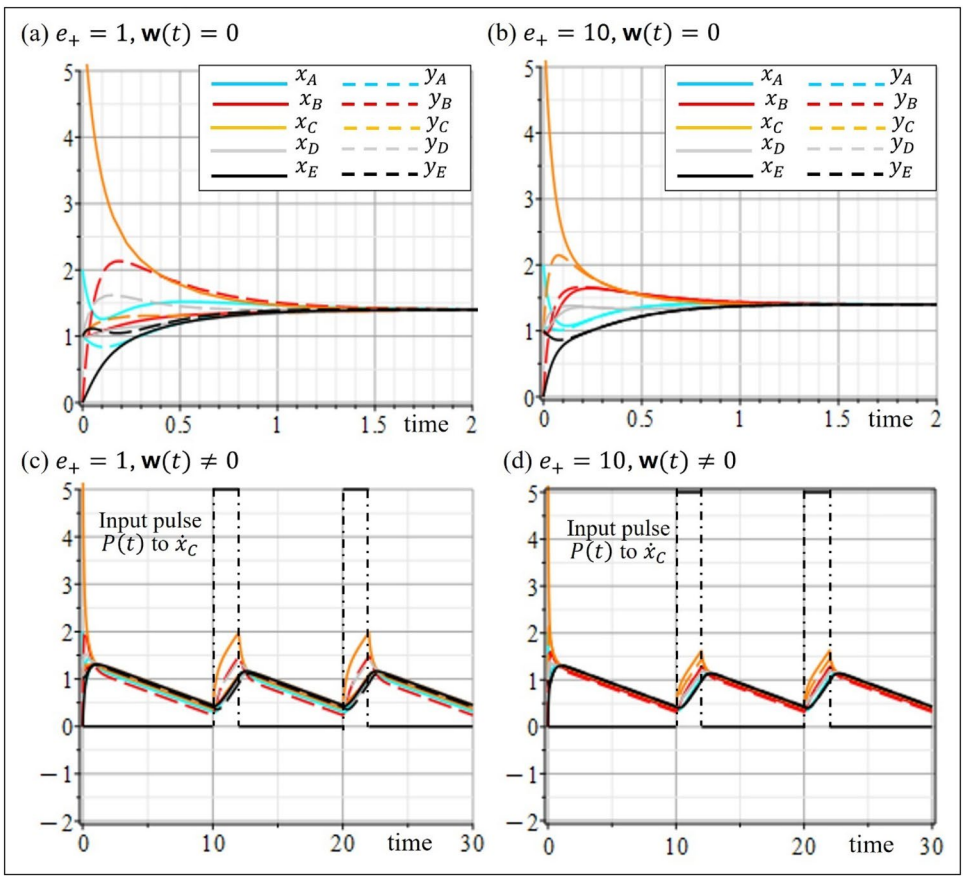



**Fig. 9** The influence of the self-conversion rate on the network convergent speed. **a** Comparing with Fig. 4a, the increase of $e_+$ from 0 to 1 helps to speed up the convergence of the network to the consensus state. **b** When $e_+$ exceeds the critical value $e_+^* = 5 - \sqrt{5} \approx 2.764$, $e_+$ has no impact on the convergent speed of the network, but it still can enhance the conversion between the yin and yang components of the individual agents. **c** Comparing with Fig. 5b, the increase of $e_+$ from 0 to 1 helps to speed up the convergence of the network to the environmental command $\mathbf{w}(t)$. **d** Further increasing $e_+$ from 1 to 10 makes the network more precisely track the environmental command $\mathbf{w}(t)$



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$$e_+^* = \sigma''_{2,3} - \sigma''_{4,5} = a_+/2 - \sqrt{5}c_+/2. \tag{35}$$

when $e_+ > e_+^*$, the influence of the self-generation mechanism on the convergence of the network $\mathbf{A}'''$ becomes saturated.

The numerical verification of the self-generation mechanism is shown in Fig. 9. The parameters used in the simulation are the same as those used in Fig. 4a, which are substituted into Eq. (35) to yield the critical self-generation rate $e_+^* = 5 - \sqrt{5} \approx 2.764$. Two self-generation rates: $e_+ = 1$and $e_+ = 10$ are considered in the numerical verification, where the latter case belongs to the saturation region. Referring to Fig. 9, the influence of the self-generation rate $e_+ = e + f$ on the convergence of network $\mathbf{A}'''$ can be divided into two regions:

- $e_+ \le e_+^*$: In the unsaturated region, the self-generation rate $e_+$ not only accelerates the balance between the Yin and Yang components of individual agents but also accelerates the balance between different agents by comparing Fig. 4a with Fig. 9a. These two figures form a control group with the former corresponding to $e_+ = 0$ and the latter corresponding to $e_+ = 1$. We can see that it takes 8 units of time for the network without self-generation to converge to the equilibrium state (see Fig. 4a), while the convergent time of Fig. 9a only takes 1.5 units of time. Figure 5b and Fig. 9c form another control group. The former is the response of the five-agent network to the environmental input $\mathbf{w}(t)$ without self-generation, and the latter is the response of the network to the environmental input with self-generation rate $e_+ = 1$. The differences between agents can be clearly identified from Fig. 5b, but in Fig. 9c, except for the component $x_C$, all components have been combined uniformly to follow the change of the environmental input $\mathbf{w}(t)$.

$e_+ > e_+^*$: After entering the saturation region, the self-generation effect only affects the Yin-Yang dynamics of individual agents, and the influence on the dynamics across different agents has reached saturation. This point can be confirmed by comparing Fig. 9a with b, and comparing Fig. 9c with d. As the self-generation rate $e_+$ increases from 1 to 10, the same color's solid lines and dotted lines (representing the same agent's Yin-Yang dynamics) merge faster so that the ten curves quickly merge into five curves. However, the five merged curves do not accelerate their convergence to the consensus value by increasing the self-generation rate beyond $e_+$. The comparison between Fig. 9a and b shows that there is almost no difference in the time required to converge to the consensus state. This result indicates that in the saturation region, the self-generation rate $e_+$ does not influence the convergence between different agents. Through the above analysis, we find that the three five-agent networks, namely the ordinary five-agent network $\mathbf{A}$, the complex five-agent network with type-II interaction $\mathbf{A}''$, and the complex five-agent network with self-generation $\mathbf{A}'''$, form a cyclic relationship, as shown in Fig. 10. If we start from the ordinary five-agent network $\mathbf{A}$ and separate each agent into Yin and Yang components, we get the complex five-agent network $\mathbf{A}''$. This process is called the Yin-Yang separation. If we add the self-generation mechanism to the complex five-agent network $\mathbf{A}''$ to promote the mutual conversion between Yin and Yang components, we obtain the network $\mathbf{A}'''$. This process is called the Yin-Yang interchange, through which the complex five-agent network $\mathbf{A}'''$ achieves the Yin-Yang balance and returns to the ordinary five-agent network $\mathbf{A}$ in which Yin and Yang are indistinguishable.

The cyclic relationship between the three five-agent networks can be explained through their eigenvalues as list in Table 1 and Fig. 8. The ten components of $\mathbf{X}'''(t)$ solved from Eq. (34) can be expressed in terms of the eigenvalue $\lambda'''$ as

$$\begin{aligned} x_i'''(t) &= A_i e^{\lambda_1''' t} + B_i e^{\lambda_{10}''' t} + C_i e^{\sigma_{2,3}''' t}\sin\left(\omega_{2,3}''' t + \theta_i\right) + D_i e^{\sigma_{4,5}''' t}\sin\left(\omega_{4,5}''' t + \varphi_i\right) \\ &+ E_i e^{\sigma_{6,7}''' t}\sin\left(\omega_{6,7}''' t + \psi_i\right) + F_i e^{\sigma_{8,9}''' t}\sin\left(\omega_{8,9}''' t + \phi_i\right), i = 1, 2, \cdots, 10 \\ &= A_i e^{\lambda_1'' t} + B_i e^{(\lambda_{10}'' - e_+)t} + C_i e^{(\sigma_{2,3}'' - e_+)t}\sin\left(\omega_{2,3}'' t + \theta_i\right) + D_i e^{\sigma_{4,5}'' t}\sin\left(\omega_{4,5}'' t + \varphi_i\right) \\ &+ E_i e^{(\sigma_{6,7}'' - e_+)t}\sin\left(\omega_{6,7}'' t + \psi_i\right) + F_i e^{\sigma_{8,9}'' t}\sin\left(\omega_{8,9}'' t + \phi_i\right), i = 1, 2, \cdots, 10, \end{aligned} \tag{36}$$

where the relationship between eigenvalues $\lambda'''$ and $\lambda''$ from Fig. 8 has been used. When the self-generation rate $e_+$ is much greater than eigenvalue $\lambda''$, we have $e^{(\lambda_{10}'' - e_+)t} \to 0$, $e^{(\sigma_{2,3}'' - e_+)t} \to 0$, $e^{(\sigma_{6,7}'' - e_+)t} \to 0$, and Eq. (36) reduces to



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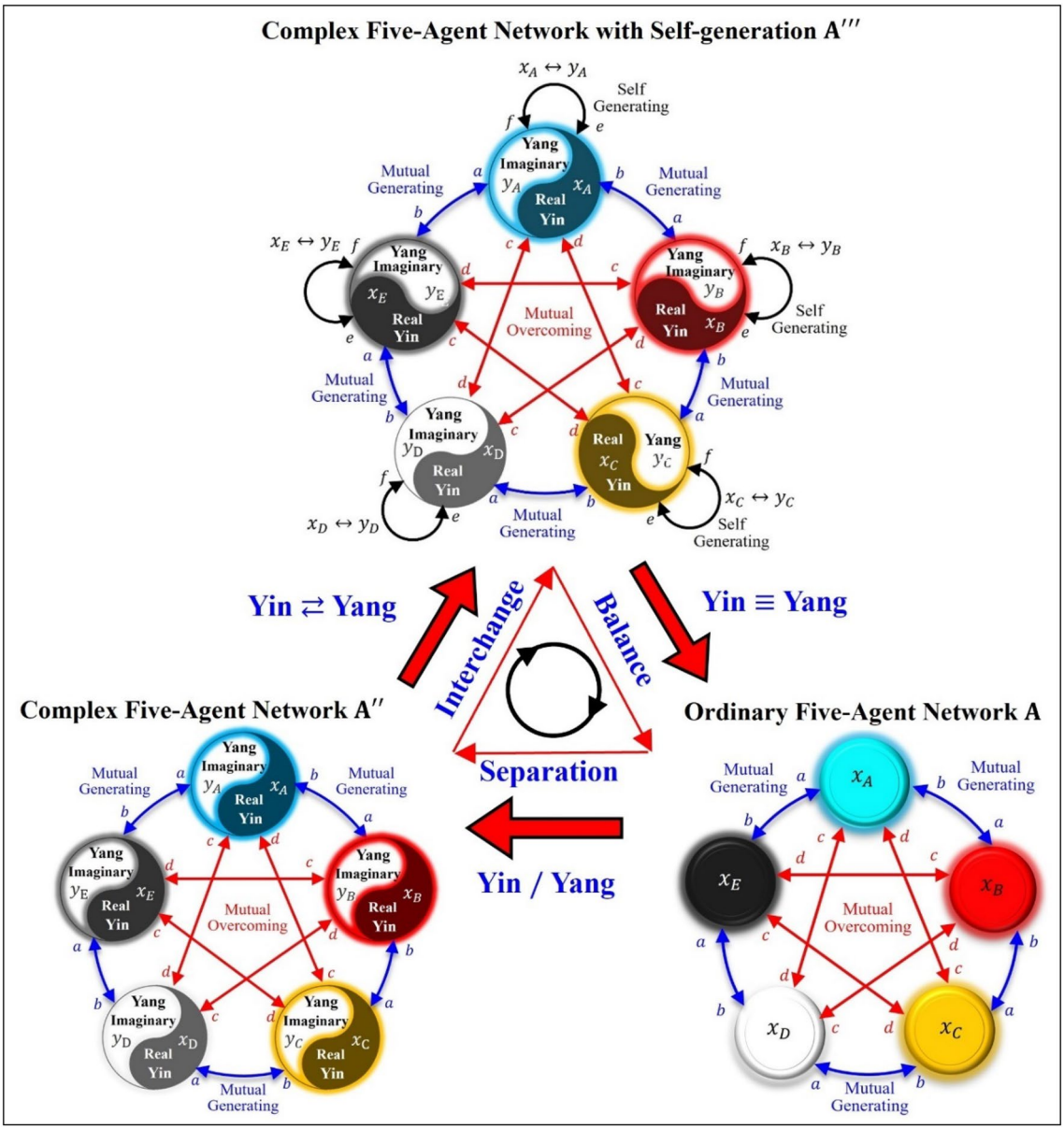



**Fig. 10** The cyclic relationship of the three five-agent networks of TCM. The ordinary five-agent network $\mathbf{A}$ is in Yin-Yang unity. The external disturbance causes each agent's Yin and Yang components to lose balance and separate from each other, leading to the complex five-agent network $\mathbf{A}''$. The self-generation mechanism is then added to the network $\mathbf{A}''$ to yield the network $\mathbf{A}'''$. The mutual exchange of Yin and Yang components in $\mathbf{A}'''$ promotes their balance and makes the system return to the ordinary five-agent network $\mathbf{A}$, in which Yin and Yang are unified again

$$\begin{aligned}\lim_{e_+\to\infty} x_i'''(t) &= A_i e^{\lambda_1'' t} + D_i e^{\sigma_{4,5}'' t}\sin\left(\omega_{4,5}'' t + \varphi_i\right) + F_i e^{\sigma_{8,9}'' t}\sin\left(\omega_{8,9}'' t + \phi_i\right)\\ &= A_i e^{\lambda_1 t} + D_i e^{\sigma_{2,3} t}\sin\left(\omega_{2,3} t + \varphi_i\right) + F_i e^{\sigma_{4,5} t}\sin\left(\omega_{4,5} t + \phi_i\right)\\ &= x_i(t),\end{aligned} \tag{37}$$

where the relationship between eigenvalues $\lambda''$ and $\lambda$ from Fig. 8 has been used. We note that $x_i(t)$ is the component of $\mathbf{X}(t)$ solved from the ordinary five-agent network (22), wherein there is no distinction between Yin and Yang attributes. The results of Eq. (36) and Eq. (37) confirms the cyclic relations that the complex five-agent network $\mathbf{A}'''$ restores to the



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ordinary five-agent network $\mathbf{A}$ through the processes of Yin-Yang exchange and Yin-Yang balance provided by the self-generation mechanism, as shown in Fig. 10.

## 6 Conclusions

This article applies the mathematical interpretation given to the Yin-Yang philosophy and the Wuxing network to provide a mathematical platform for describing the operation of TCM. This mathematical platform launches a new line of research that allows us to study TCM in a quantitative and computable way. Using this mathematical platform, we can verify many philosophical views of TCM, including the stability, balance and energy conservation of the internal organs, as well as the feedback mechanism and the automatic compensation in the human body. The potential applications of the proposed model include the analysis of organ imbalances and the predictions of the human body's response to the environmental changes and the medical treatments. The hybrid cooperative and competitive interactions in the Yin-Yang five-agent network is expected to be useful for simulating the antagonistic relationship between cancer cells and immune cells [27] and the cooperative relationship between homogeneous cells.

Nevertheless, the mathematical model established so far is limited to linear form, which only applies to organs operating near the equilibrium point. To gain concrete medical correspondence, this model has to be further infused with the clinical experience and knowledge of TCM. The most obvious example is the six parameters used in the linear model: the bidirectional rates of mutual generation $a$ and $b$, the bidirectional rates of mutual competition $c$ and $d$, and the bidirectional rates of self-generation $e$ and $f$. In the text, we assume that these six parameters are given in advance, but in fact their values must be determined from experimental data. For the same human body but under different physiological conditions, the measured six network parameters may also be different. Only network parameters determined by physiological data can be legitimately used in the stability condition (19) to determine whether the human body is in a healthy state that can balance itself. Modern system identification technology [28, 29] can help us estimate the 6 network parameters in Eq. (34) by minimizing the least-square error (Kalman filter) between the predicted states and measured states of $\mathbf{X}'''(t)$. An alternative approach is to search for the most fitted parameters such that a specified cost function is minimized by genetic algorithm [30] or Monte Carlo Simulation [31]. Considering that TCM is an empirical medicine whose diagnosis and treatment often have linguistic ambiguities under uncertain environment, it may be more practical to express the parameters with fuzzy sets [32–34], instead of expressing them with precise numbers,

Once the network parameters are determined, we can analyze the impact of environmental input $\mathbf{w}(t)$ and medical treatment $\mathbf{u}_{\text{ext}}\left(\mathbf{X}''\right)$ on the human body through Eq. (25). What is more meaningful is that before the actual medical treatment is carried out, we can predict in advance the impact of the medical behavior $\mathbf{u}_{\text{ext}}\left(\mathbf{X}''\right)$ on the human body by the numerical simulation of Eq. (25), thereby giving a trustable pre-medical assessment.

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**Data availability** All data generated or analyzed during this study are included in this published article and its supplementary information files.

## Declarations

**Competing interests** The authors declare no competing interests.

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## References

1. Maciocia G. *The Foundations of Chinese Medicine*. Elsevier, 2005.
2. Porkert M. *The Theoretical Foundations of Chinese Medicine*. MIT Press, 1974.
3. Bellomo N, From EM, Herbert A. Simon's legacy to the evolutionary artificial world with heterogeneous collective behaviors. Math Models Methods Appl Sci. 2024;34:145–80. https://doi.org/10.1142/S0218202524400049.
4. Verma H, Mishra VN, Mathur P. Effectiveness of lock down to curtail the spread of corona virus: a mathematical model. ISA Trans. 2022;124:124–34. https://doi.org/10.1016/j.isatra.2021.01.033.
5. Nath BJ, Dehingia K, Mishra VN, Chu YM, Sarmah HK. Mathematical analysis of a within-host model of SARS-CoV-2. Adv Differ Equ. 2021;2021:113. https://doi.org/10.1186/s13662-021-03276-1.
6. Yang CD. Discovering Bohr's Yin-Yang diagram in quantum tunneling dynamics. Physics. 2024;6:964–89. https://doi.org/10.3390/physics6030059.
7. Yang CD. Discovering golden ratio in the world's first five-agent network in ancient China. Sci Rep. 2023;13:13531. https://doi.org/10.1038/s41598-023-46071-6.
8. Zhu M. The Medical Classic of the Yellow Emperor. Beijing: Foreign Language Press; 2001.
9. Hou LX. The true means and development of Five Elements theory. J Yunnan Univ Tradit Chin Med. 2009;32:53–8.
10. Mesbahi M, Egerstedt M. Graph Theoretic Methods in Multiagent Networks. Princeton University Press; 2010. https://doi.org/10.1515/9781400835355.
11. Wooldridge M. *An Introduction to Multi-Agent Systems*. John Wiley & Sons, 2002.
12. Zhang ZQ, Ren JX. Exploration of the scientific connotation of the Five Elements doctrine. Chin J Basic Med in Tradit Chin Med. 2007;13:243–8.
13. Fang T. Yin Yang: a new perspective on culture. Manag Organ Rev. 2011;8:25–50. https://doi.org/10.1111/j.1740-8784.2011.00221.x.
14. Mahdihassan S. Comparing Yin-Yang, the Chinese symbol of creation, with Ouroboros of Greek alchemy. Am J Chin Med. 1989;17:95–8.
15. Plotnitsky A. Neils Bohr and Complementarity: An Introduction. New York: Springer; 2013.
16. Ribeiro C. Bohr's complementarity and Yin Yang: The mystery of Bohr's Coat of arms. Quantum Matter. 2015;4:277–83.
17. Home D. Bohr's philosophy of wave–particle complementarity. Resonance. 2013;18:905–16. https://doi.org/10.1007/s12045-013-0117-0.
18. Yang CD. Wave-particle duality in complex space. Ann Phys. 2005;319:444–70. https://doi.org/10.1016/j.aop.2005.04.014.
19. Yang CD. A scientific realization and verification of Yin-Yang theory: Complex-valued mechanics. Int J Nonlinear Sci Numer Simul. 2010;11:135–56. https://doi.org/10.1515/IJNSNS.2010.11.2.135.
20. Jozsa R. Complex weak values in quantum measurement. Phys Rev A. 2007;76: 044103. https://doi.org/10.1103/PhysRevA.76.044103.
21. Dressel J, Malik M, Miatto FM, Jordan AN, Boyd RW. Colloquium: Understanding quantum weak values: Basics and applications. Rev Mod Phys. 2014;86:307–16. https://doi.org/10.1103/RevModPhys.86.307.
22. Wu KD, et al. Operational resource theory of imaginarity. Phys Rev Lett. 2021;126: 090401. https://doi.org/10.1103/PhysRevLett.126.090401.
23. Xue S, et al. Quantification of resource theory of imaginarity. Quantum Inf Process. 2021;20:383. https://doi.org/10.1007/s11128-021-03324-5.
24. Wu KD, et al. Resource theory of imaginarity: Quantification and state conversion. Phys Rev A. 2021;103: 032401. https://doi.org/10.1103/PhysRevA.103.032401.
25. Chen MC, et al. Ruling out real-valued standard formalism of quantum theory. Phys Rev Lett. 2022;128: 040403. https://doi.org/10.1103/PhysRevLett.128.040403.
26. Wang Z, Zhang Y, Wu C. On the implication of Tai Chi diagram and its value in physics. Open Access Libr J. 2019;6:1–9. https://doi.org/10.4236/oalib.1105537.
27. Aylaj B. Qualitative analysis and simulation of a nonlinear integro-differential system modeling tumor-immune cells competition. Int J Biomath. 2018;8:1850104. https://doi.org/10.1142/S1793524518501048.
28. Torsten S, Stoica P. System Identification. New York: Prentice Hall; 1989.
29. Walter E., Pronzato L. *Identification of Parametric Models from Experimental Data*. Springer, 1997.
30. Sharma MK, Chaudhary S, Rathour L, Mishra VN. Modified genetic algorithm with novel crossover and mutation operator in travelling salesman problem. Sigma J Eng Nat Sci. 2024;42:1876–83. https://doi.org/10.14744/sigma.2023.00105.
31. Sharma MK, Dhiman N, Kumar S, Rathour L, Mishra VN. Neutrosophic monte carlo simulation approach for decision making in medical diagnostic process under uncertain environment. Int J Neutrosophic Sci. 2023;22:08–16. https://doi.org/10.54216/IJNS.220101.
32. Rathour L, Singh V, Sharma MK, Dhiman N, Mishra VN. A review of fuzzy logic analysis in COVID-19 pandemic and a new technique through extended hexagonal intuitionistic fuzzy number in analysis of COVID-19. Res Control Optim. 2024;17: 100498. https://doi.org/10.1016/j.rico.2024.100498.
33. Sharma MK, Bhargava AK, Kumar S, Rathour L, Mishra LN, Pandey S. Fermatean fuzzy ranking function in optimization of intuitionistic fuzzy transformation problems. Adv Math Mod Appl. 2022;7:191–204.
34. Rathour L, Singh V, Yadav H, Sharma MK, Mishra VN. A dual Hesitant Fuzzy set theoretic approach in Fuzzy reliability analysis of a Fuzzy system. Inf Sci Lett. 2024;13:433–40. https://doi.org/10.18576/isl/130219.

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